

REGIONAL MARITIME UNIVERSITY
FACULTY OF ENGINEERING AND APPLIED SCIENCES
BME/BMT/BEE/BNS/BCE/BCS/BIT 28B (LEVEL 100)
BSMA101-MATHEMATICS I
END OF FIRST SEMESTER EXAMINATION

INSTRUCTIONS: ANSWER ALL QUESTIONS FROM SECTION A. ANSWER ONE QUESTION FROM SECTION B. IN SECTION A, EACH QUESTION CARRY 3 MARKS.
TIME ALLOWED: 3 HOURS

1. The principal argument and modulus of the complex number $z = \sqrt{3} - i$ is
- A. $-\pi/6, 4$
 - B. $5\pi/6, 3$
 - C. $-\pi/3, 2$
 - D. $\pi/4, 6$

2. Which of the following is NOT a cube root of unity?
- A. $(\sqrt{3}i - 1)/2$
 - B. $(-\sqrt{3}i - 1)/2$
 - C. $(\sqrt{3}i + 1)/2$
 - D. 1

3. Evaluate the limit

$$\lim_{x \rightarrow \infty} \left(\frac{6^x + 40x}{7^x + 25x} \right) =$$

- A. 1.6
- B. 0
- C. 6/7
- D. 1

4. Given that $z_1 = 12e^{(-\pi/3)i}$ and $z_2 = 2e^{(\pi/3)i}$, find $\frac{z_1}{z_2}$.

- A. $6(\cos(-2\pi/3) + i \sin(-2\pi/3))$
- B. $24(\cos(0) + i \sin(0))$
- C. $6(\cos(-2\pi/9) + i \sin(-2\pi/9))$
- D. $24(\cos(-\pi/6) + i \sin(-\pi/6))$

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5. The 107^{th} order derivative of $f(\theta) = \cos \theta$ is
 A. $-\cos \theta$
 B. $-\sin \theta$
 C. $\cos \theta$
 D. $\sin \theta$
6. The sequence $u_n = \{x^{(1/x)}\}$ converges to
 A. 1
 B. 2
 C. -2
 D. 5
7. Convert $(-\sqrt{3} - i)$ into exponential form
 A. $2e^{-i(\pi/3)}$
 B. $2e^{-i(7\pi/6)}$
 C. $2e^{i(\pi/3)}$
 D. $2e^{-i(5\pi/6)}$
8. The sequence $u_n = \{1, -1, 1, -1, \dots\}$ is
 A. Convergent to -1
 B. Oscillatory
 C. Divergent
 D. Unbounded
9. All of the following are true except
 A. All the n^{th} roots of unity are in a geometric progression
 B. Sum of all the n^{th} roots of unity is zero
 C. Product of all the n^{th} roots of unity is n
 D. All the n^{th} roots of unity are lie on the circumference of the unit circle.
10. Find the value of $(-1 + i)^{30}$
 A. $-i$
 B. 2
 C. $0.707 + 0.707i$
 D. 1

Handwritten calculations for question 10:

$$\sqrt{2} \quad \sqrt{2} = \frac{1}{\sqrt{2}}$$

$$(\sqrt{2})^{30} = (2)^{15} = 32768$$

$$-0.7 + i 0.7$$

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Simplify $[1/(\cos 7\theta + i \sin 7\theta)] + \{-\cos 7\theta + i \sin 7\theta\}$

- A. $-\sin 7\theta$
- B. $-\cos 7\theta$
- C. $(\sin 7\theta + i \cos 7\theta)^7$
- D. 0

$$\frac{1}{\cos 7\theta + i \sin 7\theta} = \frac{1}{\cos 7\theta + i \sin 7\theta} \cdot \frac{\cos 7\theta - i \sin 7\theta}{\cos 7\theta - i \sin 7\theta} = \frac{\cos 7\theta - i \sin 7\theta}{\cos^2 7\theta + \sin^2 7\theta} = \cos 7\theta - i \sin 7\theta$$

12. Find the value of i^{211}

- A. -1
- B. i
- C. -i
- D. 1

$$i^{211} = i^{210} \cdot i = (i^4)^{52} \cdot i = 1^{52} \cdot i = i$$

13. The gradient of the function $y = \log|\sec \theta + \tan \theta|$ at $\theta = 5\pi/6$ is

- A. $1/2$
- B. $\sqrt{3}/2$
- C. 2
- D. $(-2\sqrt{3})/3$

$$\frac{\sec \theta \tan \theta + \sec^2 \theta}{\sec^2 \theta} = \frac{\tan \theta + \sec \theta}{\sec \theta} = \frac{1 + \sqrt{3}}{2}$$

14. What is the number c that satisfies the conclusion of the Mean Value Theorem for the function $f(x) = 1/e^x$, on the interval $[0, 2]$:

- A. 0.698
- B. 1.638
- C. 1.386
- D. 0.839

15. The parametric equations of a cycloid are $x = 4(\theta - \sin \theta)$ and $y = 4(1 - \cos \theta)$. Find dy/dx at $\theta = \pi/4$.

- A. $2\sqrt{2} - 1$
- B. 2
- C. $2\sqrt{2}$
- D. $\sqrt{2} + 1$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{4 \sin \theta}{4(1 - \cos \theta)} = \frac{\sin \theta}{1 - \cos \theta} = \frac{1/2}{1 - 1/\sqrt{2}} = \frac{1}{\sqrt{2} - 1} = \frac{\sqrt{2} + 1}{2 - 1} = \sqrt{2} + 1$$

16. Given that $x = -1 + 3t$ and $y = t^2 - t$ are the parametric equations of $y = f(x)$. What is the value of $f''(x)$ at $t = -13.5$?

- A. $4/9$
- B. $-4/9$
- C. -27
- D. $2/9$

$$y = -1 + 3t$$

$$t^2 - t = -1 + 3t$$

$$t^2 - 4t + 1 = 0$$

$$y = -1 + 3t$$

$$\frac{dy}{dx} = \frac{3}{2t - 1}$$

$$\frac{d^2y}{dx^2} = \frac{2}{(2t - 1)^2}$$

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17. Compute the gradients of the tangents drawn to ellipse $(x^2/9) + (y^2/25) = 3$ at the point
- A. ± 1.800
 B. ± 1.789
 C. ± 18.12
 D. ± 2.222

$$\frac{x^2}{9} + \frac{y^2}{25} = 3$$

18. Differentiate $y = [e^x(\sec x)]/(\log 3x)$

- A. $y' = \frac{e^x(\sec x)}{\log 3x} \left(1 + \tan x - \frac{1}{\log 3x}\right)$
 B. $y' = \frac{e^x(\sec x)}{\log 3x} \left(3 + \sec x - \frac{1}{3 \log x}\right)$
 C. $y' = \frac{e^x(\sec x)}{\log 3x} \left(-3 + \sec x - \frac{1}{\log 3x}\right)$
 D. $y' = \frac{e^x(\sec x)}{\log 3x} \left(1 + \tan x - \frac{1}{3x}\right)$

$$\frac{e^x(\sec x) \cdot 3x + e^x(\sec x) \cdot 1}{(\log 3x)^2} = \frac{e^x(\sec x)(3x + 1)}{(\log 3x)^2}$$

19. Given that $y = (x-1)^{1/x}$, the first order derivative, dy/dx , when $x = 2$ is

- A. 1.000
 B. 1.500
 C. 1.000
 D. 0.500

$$(x-1)^{1/x}$$

$$(x-1)^{1/x}$$

$$(x-1)^{1/x}$$

20. Consider the function $f(x) = \begin{cases} \frac{|x-1|}{x-1} & x \neq 1 \\ 2 & x = 1 \end{cases}$

- A. It has a limit of 1 as x approaches 1
 B. $f(1) = 1$
 C. $f(5) = f(-8) = 1$
 D. $f(x)$ is continuous at $x = 1$

21. Which of the following is the derivative of $\operatorname{cosec}^{-1}(ax)$?

- A. $(x^2\sqrt{a^2x^2 - 1})^{-1}$
 B. $(x\sqrt{a^2x^2 - 1})^{-1}$
 C. $(x\sqrt{ax - 1})^{-1}$
 D. $(-x\sqrt{a^2x^2 - 1})^{-1}$

The Taylor series for $f(x) = x \sin(x^2)$ about zero is

- A. $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (x)^{4n+2}$
- B. $\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)!} (x)^{4n+1}$
- C. $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (x)^{4n+3}$
- D. $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+3)!} (x)^{3n-1}$

23. You have 50 meters of fencing with which to fence off a rectangular backyard garden. One side of the garden is along your building and so does not need fencing. Find the maximum area possible of your garden.

- A. 625 square meters
- B. 400 square meters
- C. 312.5 square meters
- D. 300 square meters

$$2x + y = 50$$

$$A = xy$$

$$25 \times 12.5 = 312.5$$

$$\frac{dA}{dx} = x(50 - 2x)$$

$$50x - 2x^2$$

24. Find the maximum and minimum turning points of the function $y = 3 + 3x^2 - x^3$.

- A. minimum (0, 3), maximum (2, 7)
- B. minimum (1, 2), maximum (3, 8)
- C. minimum (-1, 2), maximum (1, 5)
- D. minimum (3, 1), maximum (1, 6)

$$6x - 3x^2$$

25. All of the following are true except

- A. $\cosh^2 x - \sinh^2 x = 1$
- B. $\cosh 2x = \cosh^2 x + \sinh^2 x$
- C. $\sinh x = (e^x - e^{-x})/2$
- D. $\cot^2 x = 1 + \operatorname{cosec}^2 x$

26. Find the value of $x^2 - 2x + 5$ when $x = -2 + i$

- A. $14 - 2i$
- B. $12 - 6i$
- C. $10 - 2i$
- D. $14 + 2i$

$$(-2+i)^2 - 2(-2+i) + 5$$

$$4 - 2i - 2i - 1 + 4 - 2i + 5$$

$$10 - 6i$$

27. Convert the complex number to polar form $z = -2 - 2i$

- A. $2\sqrt{2}(\cos 135^\circ + i \sin 135^\circ)$
- B. $2\sqrt{2}(\cos 225^\circ + i \sin 225^\circ)$
- C. $2\sqrt{2}(\cos -45^\circ + i \sin -45^\circ)$
- D. $2\sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$

$$2\sqrt{2}$$

28. Evaluate the derivative of the function

$$g(x) = \sqrt{\frac{1-x}{x+1}}$$

- A. $1/[(1-x)^{-1/2}(x+1)^{3/2}]$
- B. $-1/[(1-x)^{1/2}(x+1)^{-3/2}]$
- C. $1/[(1-x)^2(x+1)^{3/2}]$
- D. $-1/[(1-x)^{1/2}(x+1)^{3/2}]$

$$\frac{dy}{dx} = \frac{1}{2} \cdot \frac{\sqrt{1-x}}{\sqrt{x+1}} \cdot \frac{1}{\sqrt{1-x}} \cdot \frac{1}{\sqrt{x+1}}$$

SECTION B
(ANSWER ONLY ONE QUESTION)

- a. Solve the equation $z^4 = i$
b. Find the derivative of $y = \sin(1/\cos x)$

$-0 \quad i \quad \sqrt[4]{i}$
 $\frac{\pi}{2}$
 $\frac{\pi}{2}$

[10]

[6]

2. a. Sketch the graph of the function $f(x) = 3x^4 - 4x^3$
b. Test for convergence or divergence of the series

$$\sum_{n=0}^{\infty} \frac{1}{n^{1.001}}$$

[10]

[6]

3. a. Provide the derivative of the following function

$$y = \frac{(x-1)^3(x^{3x})e^{x^2}}{\sqrt[3]{(x^4+1)^5}} \quad \frac{dy}{dx} = (x-1)^3$$

[10]

- b. Show that $\left| \frac{z-1}{\bar{z}-1} \right| = 1$ (z is a complex number)

[6]

M.A.

1.6 0.1

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